

**Dr. A. Q. Khan Institute of Computer Science and
Information Technology**

Department of Computer Engineering

VISUAL PROGRAMMING

Nexus Learning System

Project Status Report

Submitted To

SIR UZAIR HASSAN

Submitted By

Muhammed Talha	ROLL NO# 232201054
Usama Bilal	ROLL NO# 232201097
Muhammad Abubakar	ROLL NO# 232201079

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**Dr. A. Q. Khan Institute of Computer Science and Information
Technology**

IST Campus Kahuta

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Contents

1	Project Overview	2
2	Technology Stack	2
3	Build Status	2
4	Project Structure	2
4.1	Nexus.Web	3
4.2	Nexus.Business	3
4.3	Nexus.Data	3
5	Implemented Fixes	3
5.1	Role Management Controller Fix	3
5.2	Attendance Module Enhancement	4
5.3	Connection String Resolution Improvement	4
5.4	Static Asset Improvements	4
6	Important Files	4
7	Known Issues and Diagnostics	5
7.1	Application Control Policy	5
7.2	Port Conflict	5
7.3	Duplicate Migrations	5
7.4	Temporary Files	5
8	Application Execution Procedure	6
8.1	Build Project	6
8.2	Apply Database Migrations	6
8.3	Run Application	6
9	Troubleshooting Commands	6
10	Recommended Actions	6
11	Security Considerations	7
12	Conclusion	7

1 Project Overview

The Nexus Learning System is an ASP.NET Core based Learning Management System developed using a three-tier architecture. The solution consists of three projects:

- **Nexus.Web** – Presentation Layer (MVC/Razor Pages)
- **Nexus.Business** – Business Logic Layer
- **Nexus.Data** – Data Access Layer using Entity Framework Core

Repository Location:

C:\Users\JBS\Downloads\NexusLearningSys

2 Technology Stack

- ASP.NET Core 9.0
- C
- Entity Framework Core
- SQL Server / LocalDB
- ASP.NET Identity
- Bootstrap
- Razor Pages and MVC
- HTML5, CSS3, JavaScript

3 Build Status

A full solution build was executed successfully.

- Build Result: Successful
- Compile Errors: None
- Target Framework: .NET 9.0

4 Project Structure

The project follows a layered architecture:

4.1 Nexus.Web

Responsible for:

- User Interface
- MVC Controllers
- Razor Pages
- Authentication and Authorization
- Static Files

4.2 Nexus.Business

Responsible for:

- Business Rules
- Services
- Application Logic

4.3 Nexus.Data

Responsible for:

- Database Access
- Entity Framework Core
- Migrations
- Repository Pattern

5 Implemented Fixes

5.1 Role Management Controller Fix

File Modified:

`Nexus.Web/Controllers/RoleManagementController.cs`

Changes:

- Removed `Task.WhenAll()`
- Replaced parallel operations with sequential `await`
- Prevented Entity Framework Core `DbContext` concurrency errors

5.2 Attendance Module Enhancement

Files Modified:

`Nexus.Web/Controllers/AttendanceController.cs`

`Nexus.Web/Views/Attendance/Index.cshtml`

Changes:

- Added Attendance Index action
- Added Attendance Index view
- Enabled Teacher Attendance navigation workflow

5.3 Connection String Resolution Improvement

File Modified:

`Nexus.Data/DependencyInjection/ServiceCollectionExtensions.cs`

Changes:

- Added fallback configuration loading
- Reads `Nexus.Web/appsettings.json` when required
- Improved startup reliability

5.4 Static Asset Improvements

File Modified:

`Nexus.Web/Views/Shared/_Layout.cshtml`

Changes:

- Added `asp-append-version="true"`
- Improved cache busting
- Ensured proper asset loading

6 Important Files

- `Program.cs`
- `ApplicationDbContext.cs`
- `ServiceCollectionExtensions.cs`
- `RoleManagementController.cs`
- `AttendanceController.cs`
- `DocumentRepositoryController.cs`
- `_Layout.cshtml`
- `site.css`

7 Known Issues and Diagnostics

7.1 Application Control Policy

The operating system blocks execution of generated application binaries.

Observed Error:

```
An Application Control policy has blocked this file
```

Possible Causes:

- Smart App Control
- Windows Defender Application Control (WDAC)
- AppLocker

Workaround:

```
dotnet Nexus.Web.dll
```

7.2 Port Conflict

Port 7288 was already occupied by an existing process.

Resolution:

```
netstat -ano | findstr :7288
taskkill /PID <PID> /F
```

7.3 Duplicate Migrations

Potential duplicate migration files:

```
20260602183824_TempFixNotifications.cs
20260603_AddNotifications.cs
```

Recommendation:

- Review both migration files
- Keep only the correct migration
- Remove duplicates before deployment

7.4 Temporary Files

Suspicious file detected:

```
tmp/DocumentRepositoryController.new.cs
```

Recommendation:

- Remove if not required
- Prevent build conflicts
- Avoid Razor parsing issues

8 Application Execution Procedure

8.1 Build Project

```
dotnet build
```

8.2 Apply Database Migrations

```
dotnet ef database update  
--project Nexus.Data  
--startup-project Nexus.Web
```

8.3 Run Application

```
dotnet run --project Nexus.Web  
--urls "https://localhost:7288;http://localhost:5288"
```

Alternative:

```
dotnet Nexus.Web.dll
```

9 Troubleshooting Commands

Free Port:

```
netstat -ano | findstr :7288  
taskkill /PID <PID> /F
```

Clean Build:

```
dotnet clean  
dotnet build
```

Delete Build Artifacts:

```
rm -Recurse -Force Nexus.Web\bin  
rm -Recurse -Force Nexus.Web\obj
```

10 Recommended Actions

1. Remove temporary controller files.
2. Resolve duplicate migration files.
3. Verify database migrations.
4. Move project outside the Downloads folder.
5. Run using DLL if executable execution is blocked.
6. Verify Attendance and Teacher workflows.
7. Test static assets using browser developer tools.

11 Security Considerations

- Do not commit secrets or passwords to source control.
- Use LocalDB for development environments.
- Store production credentials securely.
- Follow AppLocker and organization security policies.
- Keep dependencies updated.

12 Conclusion

The Nexus Learning System has successfully compiled without any build errors and follows a structured three-tier architecture. Several important fixes have been implemented, including improvements to role management, attendance functionality, connection string handling, and static asset management.

The remaining issues are primarily related to operating system security policies, migration cleanup, and workspace maintenance. Once these items are addressed, the system will be ready for further testing, deployment preparation, and future feature enhancements.

The project demonstrates the practical implementation of Visual Programming concepts using ASP.NET Core, Entity Framework Core, MVC architecture, and modern web development technologies.